

G Parameterization of GSM8K Templates

Motivated by GSM-Symbolic (Mirzadeh et al., 2025), we construct parameterized GSM8K templates with extended variables and corrected grammatical flaws. Each template consists of a prompt with placeholders, a variable domain, and a formula determining the answer.

Template 0

- Prompt:
 $\{Z_{\text{Name}}\}$ saw a $\{Z_x\}$ -foot $\{Z_{\text{Fish}}\}$ with $\{Z_y\}$ remoras, each $\{Z_z\}$ -inches long, attached to it. What percentage of the $\{Z_{\text{Fish}}\}$'s body length is the combined length of the remoras?
- Variables:
 $Z_{\text{Name}} \in \{\text{Benny, Alex, Casey, Jordan, Taylor, Morgan, Riley, Sam, Jamie, Avery}\}$
 $Z_{\text{Fish}} \in \{\text{dolphin, whale, shark}\}$
 $Z_k \in \{2, \dots, 21\}$
 $Z_y \in \{1, \dots, 30\}$
 $Z_{\text{ratio}} \in \{4, 5, 10, 20, 25\}$
- Formula:
 $Z_y = Z_y \times 12$
 $Z_x = Z_{\text{ratio}} \times Z_k \times Z_y / 12$
 $\text{answer} = 100 / \text{ratio}$

Template 1

- Prompt:
A $\{Z_{\text{weather}}\}$ rolls in from the ocean to cover $\{Z_{\text{city}}\}$. It takes $\{Z_t\}$ minutes to cover every $\{Z_d\}$ miles of the city. If the city is $\{Z_y\}$ miles across from the oceanfront to the opposite inland edge, how many minutes will it take for the $\{Z_{\text{weather}}\}$ to cover the whole city?
- Variables:
 $Z_{\text{city}} \in \{\text{Harborview, Lakewood, Riverton}\}$
 $Z_{\text{weather}} \in \{\text{fog bank, marine layer, coastal mist}\}$
 $Z_t \in \{2, \dots, 71\}$
 $Z_d \in \{2, \dots, 13\}$
 $Z_s \in \{1, \dots, 12\}$
- Formula:
 $Z_y = Z_d \times Z_s$
 $\text{answer} = Z_s \times Z_t$

Template 2

- Prompt:
 $\{Z_{\text{name}}\}$ is playing a $\{Z_{\text{game}}\}$ during $\{Z_{\text{setting}}\}$ for a $\{Z_{\text{prize}}\}$. $\{Z_{\text{name}}\}$ is rolling a $\{Z_s\}$ -sided die. By how many percentage points is it more likely that they roll a number greater than $\{Z_{\text{target}}\}$ than that they roll two $\{Z_z\}$ numbers in a row?
- Variables:
 $Z_{\text{name}} \in \{\text{Jerry, Alex, Casey, Jordan, Taylor, Morgan}\}$
 $Z_{\text{setting}} \in \{\text{game night, casino night, board game meetup}\}$
 $Z_{\text{game}} \in \{\text{dice game}\}$
 $Z_{\text{prize}} \in \{\text{sticker, token, pin}\}$
 $Z_{\text{target}} \in \{2, 4, 6, \dots, 200\}$
 $Z_d \in \{2, 4, 5, 10, 20, 25, 50, 100\}$
 $Z_z \in \{\text{even, odd}\}$
- Formula:
 $Z_s = Z_{\text{target}} \times Z_d$
 $\text{answer} = \frac{Z_d - 1}{Z_d} \cdot 100 - 25$

Template 3

- Prompt:
A $\{Z_{\text{group}}\}$ of $\{Z_n\}$ students has various hobbies. $\{Z_{n1}\}$ like to $\{Z_{\text{hobby1}}\}$, $\{Z_{n2}\}$ like to play $\{Z_{\text{sport}}\}$, and the rest like to either $\{Z_{\text{hobby3}}\}$ or $\{Z_{\text{hobby4}}\}$. How many like to $\{Z_{\text{hobby3}}\}$ if the number that like to $\{Z_{\text{hobby4}}\}$ is $\{Z_{\text{multi}}\}$ times the number that prefer playing $\{Z_{\text{sport}}\}$?
- Variables:
 $Z_{\text{sport}} \in \{\text{basketball, soccer}\}$
 $Z_{\text{hobby1}} \in \{\text{read, paint}\}$
 $Z_{\text{hobby3}} \in \{\text{play music, garden}\}$
 $Z_{\text{hobby4}} \in \{\text{write}\}$
 $Z_{n1} \in \{2, \dots, 11\}$
 $Z_{n2} \in \{2, \dots, 11\}$
 $Z_{\text{extra}} \in \{1, \dots, 10\}$
 $Z_{\text{multi}} \in \{2, \dots, 11\}$
- Formula:
 $Z_n = Z_{n1} + (Z_{\text{multi}} + 1) \times Z_{n2} + Z_{\text{extra}}$
 $\text{answer} = Z_{\text{extra}}$

Template 4

- Prompt:
 $\{Z_{\text{name}}\}$ is popping popcorn for a snack. As the $\{Z_p\}$ of kernels heats up, the kernels start popping faster. $\{Z_n\}$ pop in the first $\{Z_s\}$ seconds of cooking, then $\{Z_{k2}\}$ times that amount in the next $\{Z_s\}$ seconds. The kernels increase to $\{Z_{k3}\}$ times the initial popping rate in the next $\{Z_s\}$ seconds, but in the final $\{Z_s\}$ seconds, the popping slows down to $\{Z_{r1}\}$ of the rate as the past $\{Z_s\}$ seconds. After $\{Z_{\text{name}}\}$ takes the $\{Z_p\}$ off the heat, a $\{Z_{r2}\}$ of the number of kernels that popped in the final $\{Z_s\}$ seconds of cooking also pop from the residual heat. How many pieces of popcorn does $\{Z_{\text{name}}\}$ have to eat?
- Variables:
 $Z_{\text{name}} \in \{\text{Garrett, Alex, Casey}\}$
 $Z_p \in \{\text{pan, pot, skillet}\}$
 $Z_{r1} \in \{\text{half, quarter}\}$
 $Z_{r2} \in \{\text{half, quarter}\}$
 $Z_n \in \{16, 32, 48, \dots, 160\}$
 $Z_s \in \{2, 3, 4\}$
 $Z_{k2} \in \{2, \dots, 11\}$
 $Z_{k3} \in \{12, \dots, 21\}$
- Formula:
 $\text{second} = Z_{k2} \cdot Z_n$
 $\text{third} = Z_{k3} \cdot Z_n$
 $\text{fourth} = \text{third} / Z_{r1\text{num}}$
 $\text{residual} = \text{third} / (Z_{r1\text{num}} \cdot Z_{r2\text{num}})$
 $\text{answer} = Z_n + \text{second} + \text{third} + \text{fourth} + \text{residual}$

Template 5

- Prompt:
 $\{Z_{\text{name}}\}$ makes $\{Z_{\text{drink}}\}$ using teaspoons of sugar and cups of water in the ratio of $\{Z_m\}:\{Z_n\}$. If she used a total of $\{Z_x\}$ teaspoons of sugar and cups of water, calculate the number of teaspoonfuls of sugar she used.
- Variables:
 $Z_{\text{name}} \in \{\text{Katy, Sophie, Emily}\}$
 $Z_{\text{drink}} \in \{\text{coffee, tea, drink}\}$
 $Z_m \in \{1, \dots, 15\}$
 $Z_n \in \{1, \dots, 15\}$
 $Z_k \in \{1, \dots, 50\}$
- Formula:
 $Z_{\text{total}} = Z_m + Z_n$
 $Z_x = Z_{\text{total}} \cdot Z_k$
 $\text{answer} = \frac{Z_m \cdot Z_x}{Z_{\text{total}}}$

Template 6

- Prompt:
 $\{Z_{\text{name}}\}$ has $\{Z_x\}$ $\{Z_{\text{area}}\}$ s of $\{Z_{\text{fruit}}\}$ s field. There are $\{Z_k\}$ $\{Z_{\text{fruit}}\}$ s per $\{Z_{\text{area}}\}$. $\{Z_{\text{name}}\}$ can harvest his $\{Z_{\text{fruit}}\}$ s every $\{Z_n\}$ months. How many $\{Z_{\text{fruit}}\}$ s can $\{Z_{\text{name}}\}$ harvest within a year?
- Variables:
 $Z_{\text{name}} \in \{\text{John, Michael, David, James, Robert}\}$
 $Z_{\text{fruit}} \in \{\text{pineapple, apple, mango}\}$
 $Z_{\text{area}} \in \{\text{hectare, square yard}\}$
 $Z_x \in \{10, 15, 20, \dots, 50\}$
 $Z_k \in \{2, \dots, 61\}$
 $Z_n \in \{2, 3, 4, 6, 12\}$
- Formula:
 $Z_{\text{cycles}} = 12 / Z_n$
 $\text{answer} = Z_x \cdot Z_k \cdot Z_{\text{cycles}}$

Template 7

- Prompt:
 $\{Z_{\text{name}}\}$ is juggling at a $\{Z_{\text{location}}\}$. $\{Z_{\text{name}}\}$ can juggle $\{Z_n\}$ balls. $\{Z_{\text{frac1}}\}$ of the balls are $\{Z_{\text{ball_type}}\}$ balls, and $\{Z_{\text{frac2}}\}$ of the $\{Z_{\text{ball_type}}\}$ balls are $\{Z_{\text{color}}\}$. How many $\{Z_{\text{color}}\}$ $\{Z_{\text{ball_type}}\}$ balls are there?
- Variables:
 $Z_{\text{name}} \in \{\text{Jamie, Alex}\}$
 $Z_{\text{location}} \in \{\text{circus, street show}\}$
 $Z_{\text{ball_type}} \in \{\text{golf, tennis}\}$
 $Z_{\text{color}} \in \{\text{blue}\}$
 $Z_{\text{frac1}}, Z_{\text{frac2}} \in \{1/2, 1/3, 2/3, 1/4, 3/4, 1/5, 2/5, 3/5, 4/5, 1/6, 5/6, 1/7, 2/7, 3/7, 4/7, 5/7, 6/7, 1/8, 3/8, 5/8, 7/8, 1/9, 2/9, 4/9, 5/9, 7/9, 8/9, 1/10, 3/10, 7/10, 9/10\}$
 $Z_k \in \{1, \dots, 30\}$
- Formula:
 $Z_n = Z_k \times \text{denom}(Z_{\text{frac1}}) \times \text{denom}(Z_{\text{frac2}})$
 $\text{answer} = Z_n \times (\text{num}(Z_{\text{frac1}}) \times \text{num}(Z_{\text{frac2}})) / (\text{denom}(Z_{\text{frac1}}) \times \text{denom}(Z_{\text{frac2}}))$

Template 8

- Prompt:
 $\{Z_{\text{name}}\}$ places $\{Z_{\text{obj}}\}$ s on some $\{Z_{\text{surface}}\}$ s. Each $\{Z_{\text{surface}}\}$ can hold $\{Z_x\}$ $\{Z_{\text{obj}}\}$ s. If he has $\{Z_n\}$ $\{Z_{\text{obj}}\}$ s and $\{Z_k\}$ $\{Z_{\text{surface}}\}$ s, how many $\{Z_{\text{obj}}\}$ s won't he be able to place on the $\{Z_{\text{surface}}\}$?
- Variables:
 $Z_{\text{name}} \in \{\text{Jaime, John, Mike}\}$
 $Z_{\text{obj}} \in \{\text{egg, olive, almond}\}$
 $Z_{\text{surface}} \in \{\text{tray, plate, table}\}$
 $Z_x \in \{2, \dots, 21\}$
 $Z_k \in \{2, \dots, 21\}$
 $Z_{\text{leftover}} \in \{1, \dots, 10\}$
- Formula:
 $Z_n = Z_k \cdot Z_x + Z_{\text{leftover}}$
 $\text{answer} = Z_{\text{leftover}}$